

Unix and GitHub Commands Reference Sheet

Tips:

1. The up arrow retrieves your previous command.
2. You can auto-complete by hitting tab.
3. If a file or folder has a space in its name, use quotes when referring to it.
4. We always need to specify the path to a file unless it is in the working directory. We can also specify the path from the working directory to the file.
5. Typing `q` or `control-c` will escape most Unix commands.

Unix

- \$ `pwd` Shows the path to the present working directory.
- \$ `cd folder_name` Changes the current directory to the specified one.
- \$ `cd ..` Changes the current directory to the one containing the current one. Goes back one folder.
- \$ `cd ../../` Goes back two folders.
- \$ `ls` Lists all files and folders in the current directory that are not hidden.
- \$ `ls -a` Lists all hidden and unhidden files and folders in the current directory.
- \$ `ls -l` Lists with long format that shows permissions.
- \$ `mkdir folder_name` Creates a new folder in the current directory.
- \$ `rmdir folder_name` Deletes a folder in the current directory. Only works when the folder is empty.
- \$ `touch file_name` Creates a new blank file in the current directory or updates the modified date if the file already exists.
- \$ `mv path-to-file path-to-destination-directory` Moves a file to another folder.
- \$ `cp path-to-directory/file_name path-to-destination-directory/new_file_name` Copies a file to another directory.
- \$ `cp -r path-to-directory path-to-destination-directory` Copies a folder to another directory.
- \$ `rm file_name` Permanently deletes a file.
- \$ `rm -r folder_name` Permanently deletes a folder.
- \$ `man ls` Brings up the manual for the listed command. In this case, `ls`. Not available for all systems.
- \$ `ls --help` Brings up the help guide for the listed command. In this case, `ls`. Not available for all systems.

Git

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$ git config --global user.name "My Name"    Configures your user name.
$ git config --global user.mail "my@email.com"    Configures your email.
$ git config --global init.defaultBranch main    Changes the name of the default branch.
$ git status    Checks the status of new or modified files.
$ git add file_name    Adds a file to the staging area.
$ git add .    Adds all new or modified files to the staging area.
$ git add *    Adds all new or modified files to the staging area that do not have names that
               begin with a period (.).
$ git commit -m "message"    Commits all files in the staging area with a message.
$ git commit file_name -am "message"    Adds and commits the file to the staging area with
               a message.
$ git commit . -am "message"    Adds and commits all modified files to the staging area
               with a message.
$ git checkout branch_name    Switches branches.
$ git checkout -b new_branch_name    Creates a new branch.
$ git branch new_branch_name    Creates a new branch.
$ git branch    Shows the branches.
$ git merge branch_name    Merges current branch with the specified branch.
$ git init    Sets up git (specifically, a hidden .git folder) in the current directory.
$ git log    See a log of recent commits.
$ git log --all    See a log of all commits.
$ git log --all --graph    See a log of all commits with a basic graph.
$ git reset HEAD~1    Undoes the last commit.
$ git checkout commit_hash    View code in a prior commit.
$ git switch -c new_branch_name    Creates a new branch from a temporary one when check-
               ing out a prior commit.
$ git checkout commit_hash file_name    Restore file to the version in specified commit.
$ git reset commit_hash    Unstages to a certain commit hash from the log. The changes will
               still appear in your file(s).
$ git reset --hard commit_hash    Resets to a certain commit hash from the log. The changes
               will be removed from your file(s).
$ git config --global alias.alias_name "full_git_command"    Creates an alias or short-
               cut called alias_name for a full git command.
```

GitHub

\$ `git clone https://github.com/username/reponame.git` Clones the specified repo from the specified user to a folder in the current directory.

\$ `git remote add origin https://github.com/username/reponame.git`

\$ `git push` Pushes commits to GitHub.

\$ `git pull` Pulls all new or modified files to your local system and overwrites prior ones.

\$ `git fetch` Pulls all new or modified files to your local git directory, but does not overwrite files in your working directory.

\$ `git merge` Moves files from the git directory to your working directory.